

Appendix

The working: generated analysis output, every query, and the verification run.

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What is in this appendix

Everything here is **generated output**, not hand-written prose. If any figure in the submission, the dossier or the deck is ever questioned, the answer is in one of these files — and all of it can be regenerated from the source CSVs in one command.

```
cd part-c-release-verdict/analysis
python3 05_qa.py          # re-derives all 91 published figures and fails loudly on any drift
```

Contents

FILE	WHAT'S IN IT	ANSWERS
05_qa-output.txt	91 assertions , each computed from the raw CSV and checked against the published figure	<i>"Is this number real?" — start here</i>
02_main-output.txt	The full Part C Task-1 analysis: exposure ladder, tier calibration, the headline comparison with confidence intervals, all four causal tests, threshold-robustness sweep, and every segment cut	<i>"Show your calculations"</i>
03_supplement-output.txt	Target-metric definitions side by side, the CSAT inversion, the refund-gate test, the popup-decliner group, data-quality flags, and the economic sizing of the feature	<i>"What else did you check?"</i>
04_loose_ends-output.txt	Cohort baselines, the confound check on exposure, the stratified takers-vs-control comparison , challenge-start dates, and the plan-mix validation against Part D	<i>"Is the effect just activity level?"</i>
01_diagnostics-output.txt	The questions settled <i>before</i> the engagement tiers were chosen: challenge-start lag, the 8-lesson structure, dose-response by both measures, censoring, and coverage of course 338	<i>"How did you pick the cut?"</i>
sql/	All 8 BigQuery queries — see the table below for which produced data	<i>"How did you pull it?"</i>

The queries, and which ones are load-bearing

00 probes the real schema; **01** – **03** were written **before** it was known, against assumed column names, and were superseded rather than run — they are kept because they are the actual working, and they still carry their **△ADJUST** markers. Five queries produced every CSV the analysis reads.

QUERY	RAN	PRODUCED
00_explore_schema.sql	✓	95 rows — the real column list for both tables
01_engagement_tiers.sql	—	superseded draft (assumed schema)
02_core_metrics.sql	—	superseded draft (assumed schema)
03_segment_breakdown.sql	—	superseded draft (assumed schema)
10_explore_events.sql	✓	the 67-event catalogue, event×course counts, unsubscribe-reason counts, row samples

QUERY	RAN	PRODUCED
11_challenge_funnel.sql	✓	1 row — the whole-cohort funnel counts
12_challenge338_user_table.sql	✓	1,111 rows — one per challenge-338 starter
13_comparison_groups.sql	✓	9,956 rows — one per paying subscriber. The analysis dataset
14_unsub_timing.sql	not run	would make unsub % at 12h/24h reportable

The three numbers most likely to be challenged

CLAIM	WHERE TO FIND IT	TEST
The Challenge moved D1 by +0.5 pts, p = 0.82	02_main-output.txt §4 Test A	Two-proportion z-test, takers (n=1,111) vs exposure-matched control (n=994)
The effect is not an activity artefact	04_loose_ends-output.txt	Stratified by product activity — flat in 4 of 5 strata
27% of finishers completed it in one sitting	02_main-output.txt §5	13 of 48 finishers have <code>active_challenge_days = 1</code>

Method notes

- **Confidence intervals** are Wilson score intervals, not normal-approximation — several cells are small and several proportions sit near 0 or 1, where the normal approximation returns impossible bounds.
- **Group comparisons** are two-proportion z-tests with a pooled-variance standard error, reported as difference, z and p rather than as a bare "significant".
- **Censoring** is handled with explicit restricted denominators, stated on every table. The observation window is 14 days (2026-06-12 → 2026-06-25), so D7 is only observable for 72.5% of the cohort and **every unsubscribe rate is a floor**.
- **No regression, no model**. With no hold-out group in the data, the honest tool is a natural control — people who reached the same surface and declined it. Simpler than a model, and it does not hide its assumptions inside coefficients.
- The analysis scripts are **pure Python standard library** (no pandas or numpy on the machine they were written on), so they run anywhere with Python 3 and the two CSVs.

What this appendix cannot tell you

Whether the Challenge was *causal*. There is no hold-out group and no pre/post period anywhere in the warehouse — the release shipped to 100%. Every causal statement in the submission is an observational correction that worked only because a large "looked at it and left" group happened to exist as a natural control. Fixing that is the first recommendation in Part C, not the last.

Verification run — 91 assertions, all passing

```
== cohort shape ==
  ok subscribers in cohort          9956.0000 (doc: 9956)
  ok challenge-338 starters         1111.0000 (doc: 1111)
  ok non-takers                     8845.0000 (doc: 8845)
  ok all rows have a subscription   9956.0000 (doc: 9956)

== exposure ladder (Part C §1) ==
  ok saw popup                      3140.0000 (doc: 3140)
  ok saw popup %                    0.3154 (doc: 0.315)
  ok clicked popup                  2970.0000 (doc: 2970)
  ok viewed a challenge              2109.0000 (doc: 2109)
  ok started 338 %                  0.1116 (doc: 0.112)
  ok never reached a surface         6716.0000 (doc: 6716)
  ok never reached a surface %       0.6746 (doc: 0.675)

== inside the challenge (Part C §2) ==
  ok completed >= 1 lessons          612.0000 (doc: 612)
  ok completed >= 2 lessons          308.0000 (doc: 308)
  ok completed >= 3 lessons          158.0000 (doc: 158)
  ok completed >= 8 lessons           47.0000 (doc: 47)
  ok zero lessons %                  0.4491 (doc: 0.449)
  ok lesson1->2 survival              0.5033 (doc: 0.503)
  ok completed the course             48.0000 (doc: 48)
  ok certificate                     23.0000 (doc: 23)

== tiers + headline table (Part C §4) ==
  ok HIGH n                          158.0000 (doc: 158)
  ok LOW n                           953.0000 (doc: 953)
  ok HIGH D1                         0.6962 (doc: 0.696)
  ok LOW D1                          0.5845 (doc: 0.584)
  ok DIDNT-TAKE D1                   0.2174 (doc: 0.217)
  ok HIGH D3                         0.5696 (doc: 0.57)
  ok LOW D3                          0.3442 (doc: 0.344)
  ok DIDNT-TAKE D3                   0.1001 (doc: 0.1)
  ok HIGH D7                         0.3910 (doc: 0.391)
  ok HIGH D7 denominator             133.0000 (doc: 133)
  ok LOW D7                          0.1805 (doc: 0.18)
  ok LOW D7 denominator              737.0000 (doc: 737)
  ok DIDNT-TAKE D7                   0.0689 (doc: 0.069)
  ok DIDNT-TAKE D7 denominator       6345.0000 (doc: 6345)
  ok HIGH unsub                      0.1139 (doc: 0.114)
  ok LOW unsub                       0.1144 (doc: 0.114)
  ok DIDNT-TAKE unsub                0.1431 (doc: 0.143)

== the decisive test (Part C §5 Test A) ==
  ok takers D1                       0.6004 (doc: 0.6)
  ok control n                       994.0000 (doc: 994)
  ok control D1 (the 59.6%)           0.5956 (doc: 0.596)
  ok D1 effect (pts)                  0.0048 (doc: 0.005)
  ok D1 effect p-value                0.8231 (doc: 0.823)
  ok unsub effect (pts)               -0.0225 (doc: -0.023)
  ok unsub effect p-value             0.1189 (doc: 0.119)
  ok joined-never-started D1          0.6043 (doc: 0.604)
  ok viewed-never-joined D1           0.5929 (doc: 0.593)

== immortal time + dose-response (Test B / C) ==
  ok start on day 0 %                 0.2394 (doc: 0.239)
  ok day-0 dose effect (pts, negative) -0.0920 (doc: -0.092)
  ok day-0 dose p-value               0.0475 (doc: 0.048)

== the mechanic (Test D) ==
  ok finished in 1 day                13.0000 (doc: 13)
  ok finished in 1 day %              0.2708 (doc: 0.271)
  ok finished in <=2 days %           0.4583 (doc: 0.458)
```

ok	returned for a 2nd challenge day %	0.3676	(doc: 0.368)
ok	mean lessons per active challenge day	1.4608	(doc: 1.46)
== the metric that would have fooled the team (§6) ==			
ok	takers D1, challenge-anchored	0.3978	(doc: 0.398)
ok	cohort D1, first-app-day	0.2602	(doc: 0.26)
== CSAT inversion (§8) ==			
ok	CSAT never took	3.7373	(doc: 3.74)
ok	CSAT LOW	3.5338	(doc: 3.53)
ok	CSAT HIGH	3.3332	(doc: 3.33)
ok	CSAT finishers	3.1040	(doc: 3.1)
ok	CSAT of challenge-338 content	3.2543	(doc: 3.25)
ok	same users, all content	3.4504	(doc: 3.45)
== product-wide baselines (the thesis number) ==			
ok	never completed a lesson anywhere %	0.5021	(doc: 0.502)
ok	cohort unsubscribe %	0.1399	(doc: 0.14)
== demographics (Part A + the PM's claim) ==			
ok	male %	0.5974	(doc: 0.597)
ok	female %	0.3841	(doc: 0.384)
ok	aged 45+ (clean buckets) %	0.5776	(doc: 0.578)
ok	aged 45+ incl. legacy bucket %	0.6012	(doc: 0.601)
ok	men 55+ count	2120.0000	(doc: 2120)
ok	men 55+ %	0.2129	(doc: 0.213)
ok	men 35-54 %	0.2495	(doc: 0.249)
ok	unsub, age 18-24	0.3111	(doc: 0.311)
ok	unsub, age 25-34	0.1970	(doc: 0.197)
ok	unsub, age 35-44	0.1447	(doc: 0.145)
ok	unsub, age 45-54	0.0953	(doc: 0.095)
ok	unsub, age 55+	0.0969	(doc: 0.097)
ok	unsub, goal 'Work faster'	0.1095	(doc: 0.109)
ok	unsub, goal 'Feel more confident with AI'	0.1074	(doc: 0.107)
ok	unsub, goal 'Get a quick side hustle'	0.3723	(doc: 0.372)
== plans (Part D validation) ==			
ok	1Week share	0.1018	(doc: 0.102)
ok	4Week share incl. special	0.6459	(doc: 0.646)
ok	12Week share	0.2506	(doc: 0.251)
ok	1Week unsub	0.3462	(doc: 0.346)
ok	4Week unsub	0.1219	(doc: 0.122)
ok	12Week unsub	0.1058	(doc: 0.106)
ok	observed-mix blended net LTV	126.8996	(doc: 126.9)
== unsubscribe reasons (§8) ==			
ok	total unsubscribed	1393.0000	(doc: 1393)
ok	payment/chargeback/dispute unsubs	157.0000	(doc: 157)
ok	... as % of unsubs	0.1127	(doc: 0.113)
== event catalogue (the 'not observable here' claim) ==			
ok	event catalogue rows (file integrity)	67.0000	(doc: 67)
ok	subscription_renewed events	17.0000	(doc: 17)
ok	... per subscriber	0.0017	(doc: 0.0017)

=====

All 91 checks passed – every documented number matches the data.

Part C Task 1 — the full release analysis

§1 · THE EXPOSURE LADDER — of every paying subscriber, who ever reaches the Challenge?

step	n	% of cohort	step conv.
Paying subscribers in the cohort	9956	100.0%	—
... saw the Challenge popup	3140	31.5%	31.5%
... clicked the popup	2970	29.8%	94.6%
... viewed a Challenge page	2109	21.2%	71.0%
... joined Challenge 338	1341	13.5%	63.6%
... STARTED Challenge 338	1111	11.2%	82.8%
... completed ≥1 challenge lesson	613	6.2%	55.2%
... completed ≥3 challenge lessons	158	1.6%	25.8%
... completed the whole challenge	48	0.5%	30.4%
... took the certificate	23	0.2%	47.9%

Where the audience is lost (share of the whole paying cohort):

never reached any challenge surface : 6716 (67.5%)
reached a surface, never started 338: 2129 (21.4%)
started 338 : 1111 (11.2%)

§2 · ENGAGEMENT TIERS — definition and calibration

The Challenge ships 8 lessons (confirmed: every user flagged 'course_completed' has exactly 8 completions). So 'finished' = 8, and the day-3 content sits at lesson 3.

Tier rule (stated up front, applied everywhere below):

HIGH started 338 AND completed ≥ 3 of the 8 lessons (got past the Day-2 cliff)
LOW started 338 AND completed ≤ 2 lessons
DIDN'T TAKE never started 338

The cut is at 3 because that is where the D7 curve steps (18% → 39%) and it matches the product's own claim ('one day = one skill' → 3 lessons = a 3-day habit). §4 tests it for robustness at every other threshold.

Challenge-lesson completion histogram (n=1111 starters):

0 lessons	499	44.9%	cum 44.9%	
1 lessons	304	27.4%	cum 72.3%	
2 lessons	150	13.5%	cum 85.8%	
3 lessons	53	4.8%	cum 90.5%	
4 lessons	25	2.3%	cum 92.8%	
5 lessons	18	1.6%	cum 94.4%	
6 lessons	10	0.9%	cum 95.3%	
7 lessons	5	0.5%	cum 95.8%	
8 lessons	47	4.2%	cum 100.0%	

Lesson-by-lesson survival INSIDE the challenge (of the 1,111 who pressed start):

reached lesson	n	% of starters	survived prev. step
completed ≥ 1	612	55.1%	55.1%
completed ≥ 2	308	27.7%	50.3%
completed ≥ 3	158	14.2%	51.3%
completed ≥ 4	105	9.5%	66.5%
completed ≥ 5	80	7.2%	76.2%
completed ≥ 6	62	5.6%	77.5%
completed ≥ 7	52	4.7%	83.9%
completed ≥ 8	47	4.2%	90.4%

Tier sizes:

HIGH (3+ lessons) 158 (1.6% of subscribers)
LOW (0-2 lessons) 953 (9.6% of subscribers)
DIDN'T TAKE IT 8845 (88.8% of subscribers)

§3 · THE HEADLINE COMPARISON (retention anchored on each user's FIRST APP DAY for all three groups, so the groups are on the same clock – see §4 for why this matters)

metric	HIGH (3+ lessons)	LOW (0-2 lessons)	DIDN'T TAKE IT
D1 retention ★TARGET	69.6% [62.1%-76.3%]	58.4% [55.3%-61.5%]	21.7% [20.9%-22.6%]
D3 retention	57.0% [49.2%-64.4%]	34.4% [31.5%-37.5%]	10.0% [9.4%-10.6%]
D7 retention	39.1% [31.2%-47.6%]	18.0% [15.4%-21.0%]	6.9% [6.3%-7.5%]
Unsubscribe rate (D7 denominator)	11.4% [7.3%-17.3%] 133	11.4% [9.6%-13.6%] 737	14.3% [13.6%-15.1%] 6345
Avg lessons completed (all)	19.28	6.38	1.81
Avg active days in product	5.37	3.46	1.76
Avg CSAT (1-5) (CSAT raters)	3.33 158	3.53 845	3.74 3740

Significance vs 'didn't take it':

HIGH (3+ lessons)	D1	diff	+47.9%	z=+14.27	p=0.00e+00	***
HIGH (3+ lessons)	D3	diff	+47.0%	z=+18.83	p=0.00e+00	***
HIGH (3+ lessons)	D7	diff	+32.2%	z=+13.92	p=0.00e+00	***
HIGH (3+ lessons)	unsub	diff	-2.9%	z= -1.04	p=2.98e-01	ns
LOW (0-2 lessons)	D1	diff	+36.7%	z=+24.76	p=0.00e+00	***
LOW (0-2 lessons)	D3	diff	+24.4%	z=+21.74	p=0.00e+00	***
LOW (0-2 lessons)	D7	diff	+11.2%	z=+10.54	p=0.00e+00	***
LOW (0-2 lessons)	unsub	diff	-2.9%	z= -2.43	p=1.52e-02	*

§4 · IS THE GAP CAUSAL? – three tests

TEST A – EXPOSURE-MATCHED CONTROL.

'Didn't take it' is not one group: most of them were never offered the Challenge.
The fair control is people who DID reach the Challenge and chose not to engage.

exposure group	n	D1	D3	unsub
started 338 (takers)	1111	60.0% [57.1%-62.9%]	37.6% [34.8%-40.5%]	11.4% [9.7%-13.4%]
joined 338, never started	230	60.4% [54.0%-66.5%]	37.8% [31.8%-44.2%]	13.9% [10.0%-19.0%]
viewed 338, never joined	764	59.3% [55.8%-62.7%]	36.6% [33.3%-40.1%]	13.6% [11.4%-16.2%]
saw popup, never opened 338	1132	44.9% [42.0%-47.8%]	24.0% [21.6%-26.6%]	25.5% [23.1%-28.1%]
never reached a surface	6716	12.2% [11.4%-13.0%]	3.7% [3.2%-4.1%]	12.5% [11.7%-13.3%]

takers vs exposed-but-didn't-start: D1 diff +0.5% z=+0.22 p=0.823 ns
takers vs exposed-but-didn't-start: unsub diff -2.3% z=-1.56 p=0.119 ns

TEST B – IMMORTAL-TIME BIAS.

Only 24% of takers start on their first app day; the median taker starts on day +1.
To start on day +k you must already have survived to day k – so takers are pre-selected survivors. The clean read is the 'day-0 starters' who had no survival head start.

taker start-day lag: day0 266 (24%), day+1 342 (31%), day+2 196 (18%), day+3 or later 307 (28%)

group	n	D1	D3
takers who started on day 0	266	39.1% [33.4%-45.1%]	22.6% [17.9%-27.9%]
all takers (any start day)	1111	60.0% [57.1%-62.9%]	37.6% [34.8%-40.5%]
never took it	8845	21.7% [20.9%-22.6%]	10.0% [9.4%-10.6%]

day-0 takers vs never-took: D1 diff +17.4% z=+6.71 p=2.00e-11 ***

TEST C – A NON-TAUTOLOGICAL DOSE-RESPONSE.

'More lessons -> better retention' is partly circular: doing lesson 3 on day 3 IS retention.
The clean version: among users whose challenge activity happened on ONE day only, does a bigger DAY-0 dose predict coming back tomorrow? Nothing circular here.

n = 816 takers with all challenge activity on a single day

day-0 lessons	n	D1 next day	unsub
---------------	---	-------------	-------

0	429	55.7%	[51.0%-60.3%]	10.5%	[7.9%-13.7%]
1	248	54.4%	[48.2%-60.5%]	12.5%	[8.9%-17.2%]
2	99	50.5%	[40.8%-60.1%]	11.1%	[6.3%-18.8%]
3	21	33.3%	[17.2%-54.6%]	23.8%	[10.6%-45.1%]
4	5	0.0%	[0.0%-43.4%]	20.0%	[3.6%-62.4%]
8	13	53.8%	[29.1%-76.8%]	38.5%	[17.7%-64.5%]

>=2 vs <=1 day-0 lessons: D1 diff -9.2% z=-1.98 p=0.048 *

THRESHOLD ROBUSTNESS – the 'high' cut at every possible value:

high = >= k lessons	n high	D1 high	D1 low	D1 gap	unsub high	unsub low
1	612	61.4%	58.3%	+3.1%	12.7%	9.8%
2	308	63.3%	58.8%	+4.5%	13.0%	10.8%
3	158	69.6%	58.4%	+11.2%	11.4%	11.4%
4	105	74.3%	58.5%	+15.7%	10.5%	11.5%
5	80	78.8%	58.6%	+20.2%	10.0%	11.5%
6	62	80.6%	58.8%	+21.8%	11.3%	11.4%
7	52	76.9%	59.2%	+17.7%	11.5%	11.4%
8	47	76.6%	59.3%	+17.3%	12.8%	11.4%

§5 · THE MECHANIC CHECK – is this actually a DAILY challenge?

The hypothesis was 'one day = one skill builds a daily habit'. That only works if the content is gated to one lesson per day. Test: how many lessons do people do per active day?

Users who completed the whole 8-lesson challenge: 48 (4.3% of starters, 0.48% of subscribers)

... and how many DAYS they took to do it:

1 day(s):	13	(27.1%)	cum 27.1%	
2 day(s):	9	(18.8%)	cum 45.8%	
3 day(s):	6	(12.5%)	cum 58.3%	
4 day(s):	4	(8.3%)	cum 66.7%	
5 day(s):	5	(10.4%)	cum 77.1%	
6 day(s):	2	(4.2%)	cum 81.2%	
7 day(s):	3	(6.2%)	cum 87.5%	
8 day(s):	3	(6.2%)	cum 93.8%	
9 day(s):	2	(4.2%)	cum 97.9%	
10 day(s):	1	(2.1%)	cum 100.0%	

→ 13/48 (27%) finished the '7-day' challenge in ONE day.

→ 22/48 (46%) finished it in two days or fewer.

Lessons completed per active challenge day (all takers who did >=1 lesson):

~0 lessons/day:	63	(10.3%)
~1 lessons/day:	357	(58.3%)
~2 lessons/day:	136	(22.2%)
~3 lessons/day:	28	(4.6%)
~4 lessons/day:	14	(2.3%)
~5 lessons/day:	1	(0.2%)
~8 lessons/day:	13	(2.1%)

mean lessons per active challenge day: 1.46

Return behaviour: of takers who did >=1 lesson, how many ever came back for a 2nd challenge day?
225/612 = 36.8%

§6 · SEGMENTS

— AGE —								
segment	subs	% cohort	take rate	D1 (all)	unsub	D1 takers	D1	
non-tk								
55+	3200	32.1%	10.2%	25.4%	9.7%	60.9%		
21.4%								

45-54	2551	25.6%	10.9%	26.9%	9.5%	62.1%	
22.6%							
35-44	1887	19.0%	12.0%	27.8%	14.5%	66.1%	
22.5%							
25-34	1117	11.2%	12.4%	24.4%	19.7%	54.0%	
20.1%							
18-24	913	9.2%	10.3%	21.9%	31.1%	44.7%	
19.3%							
45+	235	2.4%	20.0%	37.0%	17.0%	59.6%	
31.4%							
(unknown)	53	0.5%	3.8%	13.2%	43.4%	100.0%	
9.8%							
— GENDER —							
segment	subs	% cohort	take rate	D1 (all)	unsub	D1 takers	D1
non-tk							

Male	5948	59.7%	11.3%	27.4%	12.5%	62.3%	
23.0%							
Female	3824	38.4%	11.0%	23.8%	16.1%	56.2%	
19.8%							
I'd rather skip this one	122	1.2%	10.7%	31.1%	6.6%	69.2%	
26.6%							
(unknown)	62	0.6%	4.8%	16.1%	38.7%	66.7%	
13.6%							
— PLAN —							
segment	subs	% cohort	take rate	D1 (all)	unsub	D1 takers	D1
non-tk							

4Week	6341	63.7%	11.2%	26.5%	12.2%	60.5%	
22.2%							
12Week	2495	25.1%	11.7%	26.4%	10.6%	61.3%	
21.8%							
1Week	1014	10.2%	9.5%	20.9%	34.6%	51.0%	
17.8%							
4Week_special	90	0.9%	12.2%	33.3%	3.3%	72.7%	
27.8%							
— WORK STATUS —							
segment	subs	% cohort	take rate	D1 (all)	unsub	D1 takers	D1
non-tk							

Full-time employee	3341	33.6%	11.6%	27.1%	11.8%	61.2%	
22.6%							
Business owner	1817	18.3%	8.8%	24.9%	9.4%	59.4%	
21.5%							
Freelancer / Self-employed	1487	14.9%	11.6%	26.1%	10.5%	63.0%	
21.2%							
Exploring options	845	8.5%	12.3%	26.0%	18.6%	55.8%	
21.9%							
Full-time worker	731	7.3%	12.3%	28.5%	28.5%	57.8%	
24.3%							
Between jobs / Career switcher	578	5.8%	13.1%	26.5%	13.8%	61.8%	
21.1%							
Student	175	1.8%	14.9%	20.0%	37.1%	53.8%	
14.1%							
— GOAL —							
segment	subs	% cohort	take rate	D1 (all)	unsub	D1 takers	D1
non-tk							

Work faster	3015	30.3%	10.0%	25.4%	10.9%	60.4%	
21.5%							
Feel more confident with AI	1992	20.0%	10.3%	25.5%	10.7%	62.6%	
21.2%							
Start my own business	1161	11.7%	14.4%	29.8%	10.9%	64.7%	
23.9%							
Earn more	830	8.3%	11.3%	27.7%	10.7%	64.9%	

23.0%						
Get a promotion or a better job	636	6.4%	11.0%	25.0%	14.0%	50.0%
21.9%						
Gain flexibility or work remotely	492	4.9%	13.4%	25.2%	27.4%	57.6%
20.2%						
Get a quick side hustle	368	3.7%	10.6%	21.2%	37.2%	51.3%
17.6%						
Unlock better income opportunitie	347	3.5%	15.3%	31.1%	25.4%	50.9%
27.6%						
Increase work productivity	185	1.9%	10.8%	24.3%	10.3%	60.0%
20.0%						

— COUNTRY —							
segment	subs	% cohort	take rate	D1 (all)	unsub	D1 takers	D1
non-tk							

US	3749	37.7%	9.8%	24.2%	12.5%	60.1%	
20.3%							
GB	1020	10.2%	10.0%	26.6%	12.6%	65.7%	
22.2%							
AU	691	6.9%	10.9%	25.6%	16.4%	60.0%	
21.4%							
CA	523	5.3%	12.4%	24.5%	16.3%	55.4%	
20.1%							
IT	375	3.8%	14.4%	26.1%	17.3%	53.7%	
21.5%							
AE	263	2.6%	14.8%	26.6%	12.2%	41.0%	
24.1%							
DE	255	2.6%	11.4%	26.7%	20.8%	79.3%	
19.9%							
FR	250	2.5%	10.4%	23.6%	14.4%	57.7%	
19.6%							
SG	243	2.4%	12.3%	30.5%	10.7%	56.7%	
26.8%							
ES	229	2.3%	14.8%	24.5%	14.0%	52.9%	
19.5%							
MX	202	2.0%	8.4%	29.2%	5.4%	88.2%	
23.8%							
CO	154	1.5%	10.4%	24.0%	9.1%	56.2%	
20.3%							

— AGE × GENDER (the PM's claim: 'our main market is 40-50 y.o. men') —					
age	Male	Female	other/NA	row total	row %

18-24	602	310	1	913	9.2%
25-34	613	493	11	1117	11.2%
35-44	1040	815	32	1887	19.0%
45-54	1444	1065	42	2551	25.6%
55+	2120	1035	45	3200	32.1%
TOTAL	5819	3718	131	9956	
	58.4%	37.3%			

Men aged 35-54 : 2484 (24.9% of paying subscribers)
 Everyone aged 35-54 : 4438 (44.6%)
 All men : 5819 (58.4%)
 All women : 3718 (37.3%)
 Aged 45+ : 5751 (57.8%)

§7 · UNSUBSCRIBES

total unsubscribed in cohort: 1393 / 9956 = 14.0%

plan	subs	unsub	rate

4Week	6341	773	12.2% [11.4%–13.0%]
12Week	2495	264	10.6% [9.4%–11.8%]
1Week	1014	351	34.6% [31.8%–37.6%]
4Week_special	90	3	3.3% [1.1%–9.3%]

tier	n	unsub	rate
HIGH (3+ lessons)	158	18	11.4% [7.3%–17.3%]
LOW (0–2 lessons)	953	109	11.4% [9.6%–13.6%]
DIDN'T TAKE IT	8845	1266	14.3% [13.6%–15.1%]

Stated reason (only recorded for some unsubsribes):

(none recorded)	873	(62.7% of unsubs)
Support request	310	(22.3% of unsubs)
hard_decline	84	(6.0% of unsubs)
account_deletion	31	(2.2% of unsubs)
Mastercard Alert	30	(2.2% of unsubs)
general	22	(1.6% of unsubs)
dispute	12	(0.9% of unsubs)
paypal_refund	10	(0.7% of unsubs)
Visa CDRN	7	(0.5% of unsubs)
fraud	6	(0.4% of unsubs)
Merchanto visa	4	(0.3% of unsubs)
payment_refunded	3	(0.2% of unsubs)
fraud_reported	1	(0.1% of unsubs)

→ payment-failure / chargeback / refund-dispute reasons: 157 (11.3% of unsubs, 1.6% of the cohort)

Do challenge takers unsubscribe less? (raw, uncontrolled)

takers	127/1111	= 11.4%	[9.7%–13.4%]
non-takers	1266/8845	= 14.3%	[13.6%–15.1%]
diff	-2.9%	z=-2.61	p=0.009 **

Decisions settled before the tiers were chosen

subscribers: 9956 · challenge-338 takers: 1111

Q1 · lag between first app day and challenge start (days)

+ 0 days:	266	(23.9%)	cum	23.9%
+ 1 days:	342	(30.8%)	cum	54.7%
+ 2 days:	196	(17.6%)	cum	72.4%
+ 3 days:	90	(8.1%)	cum	80.5%
+ 4 days:	65	(5.9%)	cum	86.3%
+ 5 days:	58	(5.2%)	cum	91.5%
+ 6 days:	42	(3.8%)	cum	95.3%
+ 7 days:	24	(2.2%)	cum	97.5%
+ 8 days:	15	(1.4%)	cum	98.8%
+ 9 days:	3	(0.3%)	cum	99.1%
+10 days:	6	(0.5%)	cum	99.6%
+11 days:	2	(0.2%)	cum	99.8%
+12 days:	1	(0.1%)	cum	99.9%
+13 days:	1	(0.1%)	cum	100.0%

→ Only 23.9% of takers start the challenge on their FIRST app day;
the median taker starts on day +1, and 28% start on day +3 or later.
CONSEQUENCE: the two anchors are NOT interchangeable, and takers are pre-selected survivors – to start on day k you must have survived to day k (immortal-time bias).
So: use the first-seen anchor for all cross-group comparisons, and treat even that as favourable to takers. See 02_main.py §4 Test B.

Q2 · challenge lessons STARTED vs COMPLETED (the >7 spike)

completed: {0: 499, 1: 304, 2: 150, 3: 53, 4: 25, 5: 18, 6: 10, 7: 5, 8: 47}
started: {0: 1, 1: 698, 2: 234, 3: 60, 4: 32, 5: 19, 6: 12, 7: 6, 8: 49}

users with >7 completions: 47
their completed-course flag: Counter({1: 47})
their certificate flag: Counter({0: 25, 1: 22})
their active_challenge_days: {1: 13, 2: 9, 3: 5, 4: 4, 5: 5, 6: 2, 7: 3, 8: 3, 9: 2, 10: 1}
their lessons_started: {8: 47}

cross-check – completed_course flag by completion count:

0 lessons:	course_completed	0/ 499
1 lessons:	course_completed	0/ 304
2 lessons:	course_completed	0/ 150
3 lessons:	course_completed	0/ 53
4 lessons:	course_completed	0/ 25
5 lessons:	course_completed	0/ 18
6 lessons:	course_completed	0/ 10
7 lessons:	course_completed	1/ 5
8 lessons:	course_completed	47/ 47

Q3 · DOSE-RESPONSE: outcome by exact number of challenge lessons completed
(retention anchored on first_seen so it is comparable across everyone)

lessons	n		D1	D3	D7 (n obs)		unsub
0	499	58.3% [53.9%–62.6%]	33.5% [29.5%–37.7%]	18.5% [15.0%–22.7%]	(383)	9.8% [7.5%–12.7%]	
1	304	59.5% [53.9%–64.9%]	34.5% [29.4%–40.0%]	17.2% [13.0%–22.5%]	(238)	12.5% [9.2%–16.7%]	
2	150	56.7% [48.7%–64.3%]	37.3% [30.0%–45.3%]	18.1% [12.2%–26.1%]	(116)	14.7% [9.9%–21.2%]	
3	53	60.4% [46.9%–72.4%]	35.8% [24.3%–49.3%]	39.0% [25.7%–54.3%]	(41)	13.2% [6.5%–24.8%]	
4	25	60.0% [40.7%–76.6%]	68.0% [48.4%–82.8%]	31.8% [16.4%–52.7%]	(22)	12.0% [4.2%–30.0%]	
5	18	72.2% [49.1%–87.5%]	61.1% [38.6%–79.7%]	38.5% [17.7%–64.5%]	(13)	5.6% [1.0%–25.8%]	
6	10	100.0% [72.2%–100.0%]	80.0% [49.0%–94.3%]	50.0% [21.5%–78.5%]	(8)	10.0% [1.8%–40.4%]	

43.4%]	7	5		80.0% [37.6%-96.4%]	100.0% [56.6%-100.0%]	40.0% [11.8%-76.9%]	(5)		0.0% [0.0%-
	8	47		76.6% [62.8%-86.4%]	63.8% [49.5%-76.0%]	40.9% [27.7%-55.6%]	(44)		12.8% [6.0%-25.2%]

same, by ACTIVE CHALLENGE DAYS (the mechanic's own unit: 1 day = 1 skill)

lessons	n		D1	D3	D7 (n obs)		unsub

14.4%]	1	816		53.7% [50.2%-57.1%]	29.4% [26.4%-32.6%]	14.7% [12.1%-17.7%]	(614) 12.0% [10.0%-
	2	163		76.1% [69.0%-82.0%]	46.6% [39.1%-54.3%]	28.6% [21.7%-36.5%]	(140) 11.0% [7.1%-16.8%]
	3	59		76.3% [64.0%-85.3%]	74.6% [62.2%-83.9%]	36.0% [24.1%-49.9%]	(50) 8.5% [3.7%-18.4%]
	4	29		79.3% [61.6%-90.2%]	79.3% [61.6%-90.2%]	39.1% [22.2%-59.2%]	(23) 10.3% [3.6%-26.4%]
	5	21		90.5% [71.1%-97.3%]	76.2% [54.9%-89.4%]	52.4% [32.4%-71.7%]	(21) 14.3% [5.0%-34.6%]
32.4%]	6	8		62.5% [30.6%-86.3%]	100.0% [67.6%-100.0%]	71.4% [35.9%-91.8%]	(7) 0.0% [0.0%-
	7	15		86.7% [62.1%-96.3%]	73.3% [48.0%-89.1%]	80.0% [54.8%-93.0%]	(15) 0.0% [0.0%-20.4%]

=====
Q4 · observation window / censoring
=====

```

last data date: 2026-06-25
days from first_seen to last data date:
  0 days:      3
  1 days:     10
  2 days:      2
  3 days:     10
  4 days:    862
  5 days:    878
  6 days:    976
  7 days:   1167
  8 days:   1270
  9 days:   1051
 10 days:    911
 11 days:   1057
 12 days:    877
 13 days:    882
D1 observable for 9953/9956 (100.0%) of subscribers
D3 observable for 9941/9956 (99.8%) of subscribers
D7 observable for 7215/9956 (72.5%) of subscribers

```

=====
Q5 · is challenge 338 effectively THE Challenge feature?
=====

```

started_any_challenge: 1114
group_338 == took_338: 1111
took_other_challenge : 3
→ 338 covers 99.7% of all challenge starters.

```

Supplementary tests — CSAT, refund gate, data quality

§A · D1 AS THE TEAM DEFINED IT vs D1 ON A COMMON CLOCK

Brief: 'Target metric: D1 Retention (retention the day after the challenge started).'
That anchors takers on their challenge-start day. Non-takers have no such day, so the only way to compare is a common clock (first app day). Both are shown – the difference is the single biggest reason a naive read of this release looks better than it is.

measure	n	rate
TAKERS · D1 anchored on challenge-start (team defn)	1111	39.8% [36.9%–42.7%]
TAKERS · D1 anchored on first app day	1111	60.0% [57.1%–62.9%]
NON-TAKERS · D1 anchored on first app day	8845	21.7% [20.9%–22.6%]
TAKERS · D3 anchored on challenge-start	982	25.9% [23.2%–28.7%]
TAKERS · D7 anchored on challenge-start	606	14.7% [12.1%–17.7%]

§B · CSAT – does engaging with the product make people happier?

group	raters	mean CSAT	% 4-5	% 1-2
Completed the whole challenge	48	3.10	37.5%	22.9%
HIGH (3+ challenge lessons)	158	3.33	38.6%	19.6%
LOW (0-2 challenge lessons)	845	3.53	53.0%	20.6%
Never took the challenge	3740	3.74	59.8%	17.5%

CSAT given INSIDE challenge 338 specifically: n=583 raters, mean 3.25
... of 1111 starters, only 52.5% ever rated challenge content.
Same users' CSAT across ALL product content: 3.45

§C · COMPLETION AND THE REFUND GATE

Part A found (from public reviews) that finishing the plan is the money-back gate.
If true, completion should predict MORE unsubscribing, not less. Test it.

group	n	unsubscribe rate
finished the challenge (all 8 lessons)	48	12.5% [5.9%–24.7%]
started but did not finish	1063	11.4% [9.6%–13.4%]
never took it	8845	14.3% [13.6%–15.1%]

finishers vs non-finishers: diff +1.1% z=+0.24 p=0.812 ns

of the finishers, those who binged it in <=2 days: n=22, unsub 22.7% [10.1%–43.4%]
of the finishers, those who spread it over 3+ days: n=26, unsub 3.8% [0.7%–18.9%]

took the certificate: n=23, unsub 13.0% [4.5%–32.1%]

§D · THE 'SAW THE POPUP AND WALKED AWAY' GROUP – the worst cell in the dataset

group	n	D1	unsub	avg lessons
saw popup, never opened the challenge	1132	44.9% [42.0%–47.8%]	25.5% [23.1%–28.1%]	3.71
never reached any challenge surface	6716	12.2% [11.4%–13.0%]	12.5% [11.7%–13.3%]	0.81
took the challenge	1111	60.0% [57.1%–62.9%]	11.4% [9.7%–13.4%]	8.22

plan mix of 'saw popup, never opened': [('4Week', 703), ('12Week', 266), ('1Week', 144), ('4Week_special', 15)]
plan mix of the whole cohort : [('4Week', 6341), ('12Week', 2495), ('1Week', 1014), ('4Week_special', 90)]

1-Week subscribers as share of 'saw popup, never opened': 12.7% (cohort: 10.2%)

=====

\$E · DATA-QUALITY FLAGS (things I would raise before anyone acts on this)

=====

1. Two different quiz vocabularies are mixed in one cohort:
age '45-54': n=2551 unsub= 9.5% D1= 26.9%
age '55+': n=3200 unsub= 9.7% D1= 25.4%
age '45+': n=235 unsub= 17.0% D1= 37.0%
status 'Full-time employee': n=3341 unsub= 11.8% D1= 27.1%
status 'Full-time worker': n=731 unsub= 28.5% D1= 28.5%
→ 'Full-time worker' churns at 2.4x 'Full-time employee'. Same words, different funnel.
Either a different quiz version or different traffic. Cannot be pooled blindly.
2. THE BIG ONE – there is no hold-out group. Challenge starts run on every single day of the window (2026-06-12 ... 2026-06-25), so there is no pre/post and no control arm. Every causal statement below is an observational correction, not an experiment. This is the #1 thing to fix before the next release, not after it.
3. Observation window is only 14 days (2026-06-12 → 2026-06-25):
D7 is unobservable for 27.5% of the cohort;
unsubscribe is right-censored for everyone (a 4-week plan cannot even reach its first renewal inside this window), so all unsubscribe rates here are FLOOR values.
4. The popup carries no challenge_id, so 'saw the popup' is exposure to ANY challenge, not specifically 338. 338 is 99.7% of all starts, so the practical impact is small.

=====

\$F · WHAT THE FEATURE IS WORTH TODAY

=====

Reach: 1111/9956 = 11.2% of paying subscribers started it.
Activation: 612/1111 = 55.1% did even one lesson.
Completion: 48/1111 = 4.3% finished (0.48% of all subscribers).

Best causal estimate of the D1 effect (vs the exposure-matched control):
+0.5% percentage points, p=0.823 (ns) – statistically indistinguishable from zero.

Even if the effect were real and equal to the raw taker-vs-non-taker gap, at 11.2% reach the cohort-level D1 impact would be at most 0.112 x the gap.

raw gap = +38.3% → cohort-level ceiling = +4.3% pts of overall D1.

Overall cohort D1 today: 26.0%

Cohort baselines and the stratified control test

```
== cohort-level baselines (all 9,956 paying subscribers, first-app-day anchor) ==
D1: 26.0% [25.2%-26.9%]    (n=9953)
D3: 13.1% [12.5%-13.8%]    (n=9941)
D7: 8.6% [8.0%-9.3%]      (n=7215)
unsubscribed within the 14-day window: 14.0% [13.3%-14.7%]
mean lessons completed (any course): 2.53
mean active days in product: 1.98
never completed a single lesson anywhere: 50.2%
finished onboarding: 69.8%

== CONFOUND CHECK for $D: is 'saw the popup' just a proxy for time-in-product? ==
group                n  mean active days  mean days observed  unsub
took 338              1111                3.73                8.94       11.4%
saw popup, never opened 1132                2.74                8.95       25.5%
never reached a surface 6716                1.33                8.27       12.5%

Same three groups, holding ACTIVE DAYS roughly constant (users with 3-6 active days):
group                n  D1                unsub
took 338              622 69.6% [65.9%-73.1%]  11.3% [9.0%-14.0%]
saw popup, never opened 516 63.4% [59.1%-67.4%]  24.0% [20.5%-27.9%]
never reached a surface 326 58.9% [53.5%-64.1%]  19.3% [15.4%-24.0%]

== TAKERS vs EXPOSED-CONTROL, holding active days constant (the tightest test I can run) ==
active days  n takers  D1 takers  n control  D1 control  diff  p
1            97       0.0%       63       0.0%      +0.0%  1.000
2           270      46.3%      257      42.4%      +3.9%  0.370
3-4          446      64.8%      427      65.1%      -0.3%  0.924
5-7          217      82.9%      205      84.4%      -1.4%  0.689
8+           81      90.1%       42      76.2%     +13.9%  0.038

== was the Challenge live for the whole observation window? ==
2026-06-12: 39 starts
2026-06-13: 69 starts
2026-06-14: 101 starts
2026-06-15: 124 starts
2026-06-16: 70 starts
2026-06-17: 88 starts
2026-06-18: 115 starts
2026-06-19: 102 starts
2026-06-20: 84 starts
2026-06-21: 84 starts
2026-06-22: 106 starts
2026-06-23: 76 starts
2026-06-24: 38 starts
2026-06-25: 15 starts

== plan mix: real cohort vs the plans.csv assumption used in Part D ==
1Week : 10.2% (plans.csv assumes 10%)
4Week : 64.6% (plans.csv assumes 70%)
12Week: 25.1% (plans.csv assumes 20%)
```

Part A — k-modes segment clustering

```
quiz respondents in export          49,487
answered >= 6 of 10 core questions 38,071 (76.9%) <- the clustering sample

== choosing k: within-cluster Hamming cost ==
k      cost    cost/user  improvement
2  193,760      5.089      -
3  182,252      4.787      5.9%
4  177,769      4.669      2.5%
5  171,100      4.494      3.8%
6  167,414      4.397      2.2%
7  164,227      4.314      1.9%
8  159,839      4.198      2.7%

== k = 6 · cluster profiles (lift = share in cluster / share in population) ==

— cluster 0  n = 10,810 (28.4%)
  1.87x 62.3% feels_ready      Somewhat ready – but I could be behind
  1.63x 49.6% learn_claude_for Growth - I love learning in-demand skills
  1.61x 42.7% benefit         Feel more confident with AI
  1.52x 41.8% age             45-54
  1.33x 69.7% ai_experience    Good, but i feel like i'm only scratching the surfac
  1.28x 75.0% gender          Male

— cluster 3  n = 8,372 (22.0%)
  2.22x 50.3% feels_ready      A bit worried – things are changing quickly
  2.17x 61.0% career_fear      Falling behind as others move faster
  1.87x 82.0% used_claude      Not yet
  1.59x 40.4% field            Other
  1.43x 48.9% age             55+
  1.29x 44.6% benefit          Work faster

— cluster 1  n = 7,953 (20.9%)
  2.51x 45.7% feels_ready      Yes - I'm confident in my skills
  2.50x 60.6% ai_experience     Great - AI already helps me a lot
  1.71x 67.2% gender           Female
  1.66x 57.1% benefit          Work faster
  1.49x 80.1% learn_claude_for Work tasks
  1.39x 58.2% career_fear      Nothing - I see AI as an opportunity, not a threat

— cluster 2  n = 4,699 (12.3%)
  2.59x 49.8% age             35-44
  1.49x 51.4% benefit          Work faster
  1.39x 75.0% learn_claude_for Work tasks
  1.31x 76.6% gender           Male
  1.28x 71.5% used_claude      Yes
  1.17x 49.6% work_status      Full-time employee

— cluster 4  n = 3,413 (9.0%)
  3.11x 50.4% benefit          Start my own business
  2.56x 46.5% feels_ready      Not really – I need new skills fast
  2.16x 84.9% gender           Female
  2.07x 52.7% field            Other
  1.99x 60.8% learn_claude_for Growth - I love learning in-demand skills
  1.77x 48.6% age             45-54

— cluster 5  n = 2,824 (7.4%)
  4.42x 49.6% work_status      Exploring options
  2.43x 44.2% feels_ready      Not really – I need new skills fast
  2.32x 27.2% ai_experience     I haven't really tried yet
  2.22x 58.6% benefit          Feel more confident with AI
  2.04x 31.9% learn_claude_for Personal use
  2.01x 68.9% age             55+
```

SQL · 00_explore_schema

```
-- Part C · Step 0 – LEARN THE REAL SCHEMA FIRST (run this before anything else)
-- Project: persona-496908 · Tables: app_events, subscribe_events
-- The queries in 01-04 use ASSUMED column names (flagged with -- ΔADJUST). Run these
-- three first, then find/replace the assumed names with the real ones.

-- 1) What columns exist? (replace <dataset> with the real dataset id shown in the console)
SELECT table_name, column_name, data_type
FROM `persona-496908.<dataset>.INFORMATION_SCHEMA.COLUMNS`
WHERE table_name IN ('app_events', 'subscribe_events')
ORDER BY table_name, ordinal_position;

-- 2) What do app_events rows look like, and what event names exist?
SELECT * FROM `persona-496908.<dataset>.app_events` LIMIT 50;

SELECT event_name, COUNT(*) AS n          -- ΔADJUST: the event-name column may be `event`, `name`, `type`
FROM `persona-496908.<dataset>.app_events`
GROUP BY event_name ORDER BY n DESC LIMIT 100;
-- ↑ Look here for the Challenge events: something like challenge_start / challenge_day_complete /
--   lesson_complete / day_open. Note the exact strings – 01-04 depend on them.

-- 3) What does subscribe_events carry? (need: user id, subscription start, cancel/unsubscribe, plan, price)
SELECT * FROM `persona-496908.<dataset>.subscribe_events` LIMIT 50;
```

SQL · 01_engagement_tiers

```
-- Part C · Step 1 – Classify each Challenge-eligible user into an engagement tier.
-- ΔADJUST all names marked below to the real schema found in 00_explore_schema.sql.
-- Tiers: high = started AND (reached >=Day3 OR returned on >=3 of first 7 days)
--        low  = started but stalled before Day 3
--        none = eligible but never opened Day 1

DECLARE challenge_id STRING DEFAULT '338';          -- the Challenge in the brief (skills/challenges/338)

WITH ev AS (
  SELECT
    user_id,                                -- ΔADJUST
    event_name,                             -- ΔADJUST
    TIMESTAMP(event_timestamp) AS ts,        -- ΔADJUST
    DATE(TIMESTAMP(event_timestamp)) AS d,
    SAFE_CAST(JSON_VALUE(event_params, '$.challenge_id') AS STRING) AS ch -- ΔADJUST (may be a column)
  FROM `persona-496908.<dataset>.app_events`
),
challenge AS (                                -- events belonging to THIS challenge
  SELECT * FROM ev WHERE ch = challenge_id OR ch IS NULL -- drop the OR NULL once the field is confirmed
),
per_user AS (
  SELECT
    user_id,
    MIN(IF(event_name='challenge_start', ts, NULL)) AS started_at,      -- ΔADJUST event
    COUNTIF(event_name='challenge_day_complete') AS days_completed,    -- ΔADJUST
    event
  COUNT(DISTINCT IF(event_name='challenge_day_complete', d, NULL)) AS distinct_active_days,
    MIN(d) AS first_day, MAX(d) AS last_day
  FROM challenge
  GROUP BY user_id
)
SELECT
  user_id,
  started_at IS NOT NULL AS started,
  days_completed,
  distinct_active_days,
  CASE
    WHEN started_at IS NULL THEN 'none'
    WHEN days_completed >= 3 OR distinct_active_days >= 3 THEN 'high'
    ELSE 'low'
  END AS engagement_tier
FROM per_user;

-- Sanity: the tier distribution (use this histogram to calibrate the Day-3 threshold)
-- SELECT engagement_tier, COUNT(*) FROM ( <the query above> ) GROUP BY 1;
```

SQL · 02_core_metrics

```
-- Part C · Step 2 – The core table: target + secondary metrics BY engagement tier.
-- Depends on the tiering in 01. ΔADJUST names to the real schema.
-- D1/D3/D7 retention are measured relative to each user's challenge-start day.

DECLARE challenge_id STRING DEFAULT '338';

WITH tiers AS (
  -- paste/inline the SELECT from 01_engagement_tiers.sql, or save it as a view and reference it
  SELECT * FROM `persona-496908.<dataset>.v_challenge_tiers`      -- Δ create this view from 01
),
starts AS (
  SELECT user_id, DATE(MIN(TIMESTAMP(event_timestamp))) AS d0    -- ΔADJUST
  FROM `persona-496908.<dataset>.app_events`
  WHERE event_name='challenge_start'                             -- ΔADJUST
  GROUP BY user_id
),
activity AS (
  SELECT DISTINCT user_id, DATE(TIMESTAMP(event_timestamp)) AS d -- any app activity = "retained that day"
  FROM `persona-496908.<dataset>.app_events`
),
ret AS (
  SELECT s.user_id,
    MAX(a.d = DATE_ADD(s.d0, INTERVAL 1 DAY)) AS d1,
    MAX(a.d = DATE_ADD(s.d0, INTERVAL 3 DAY)) AS d3,
    MAX(a.d = DATE_ADD(s.d0, INTERVAL 7 DAY)) AS d7
  FROM starts s LEFT JOIN activity a USING(user_id)
  GROUP BY s.user_id
),
unsub AS (
  SELECT user_id, MAX(is_cancelled) AS unsubscribed             -- ΔADJUST: from subscribe_events
  FROM `persona-496908.<dataset>.subscribe_events`              -- e.g. status='cancelled' / cancel_at IS
  NOT NULL
  GROUP BY user_id
)
SELECT
  t.engagement_tier,
  COUNT(*)                                AS users,
  ROUND(AVG(CAST(r.d1 AS INT64)),3)        AS d1_retention,    -- TARGET METRIC
  ROUND(AVG(CAST(r.d3 AS INT64)),3)        AS d3_retention,
  ROUND(AVG(CAST(r.d7 AS INT64)),3)        AS d7_retention,
  ROUND(AVG(CAST(COALESCE(u.unsubscribed,false) AS INT64)),3) AS unsubscribe_rate,
  ROUND(AVG(t.days_completed)/7.0,3)       AS lesson_completion -- of the 7 days
FROM tiers t
LEFT JOIN ret r USING(user_id)
LEFT JOIN unsub u USING(user_id)
GROUP BY t.engagement_tier
ORDER BY CASE t.engagement_tier WHEN 'high' THEN 1 WHEN 'low' THEN 2 ELSE 3 END;

-- READ IT LIKE THIS: the release "worked" only if there is a smooth dose-response
-- (high < low < none on unsubscribe; high > low > none on D1/D3/D7) that survives the
-- baseline/hold-out check in 03. A high-vs-none gap alone can be pure selection.
```

SQL · 03_segment_breakdown

```
-- Part C · Step 3 – Same metrics, cut by segment (age / profession / plan) + the selection guard.
-- ΔADJUST names. Quiz answers (age, status/profession, goal) live on the user/subscribe record or
-- in early app_events params – confirm where in 00.

-- 3a) Metrics by demographic segment × engagement tier
WITH base AS (
  SELECT t.user_id, t.engagement_tier,
         s.age, s.profession, s.plan, -- ΔADJUST source of quiz demographics
         r.d1, r.unsubscribed, t.days_completed
  FROM `persona-496908.<dataset>.v_challenge_metrics_user` t -- Δ a per-user view joining 01+02
  LEFT JOIN `persona-496908.<dataset>.subscribe_events` s USING(user_id)
  LEFT JOIN `persona-496908.<dataset>.v_challenge_ret` r USING(user_id)
)
SELECT age, plan, engagement_tier,
       COUNT(*) users,
       ROUND(AVG(CAST(d1 AS INT64)),3) d1_retention,
       ROUND(AVG(CAST(unsubscribed AS INT64)),3) unsubscribe_rate
FROM base
GROUP BY age, plan, engagement_tier
ORDER BY age, plan, engagement_tier;

-- 3b) THE SELECTION GUARD – dose-response: does each EXTRA challenge day predict lower unsubscribe?
-- If the slope is smooth and monotonic, the effect is more plausibly causal than pure self-selection.
SELECT days_completed,
       COUNT(*) users,
       ROUND(AVG(CAST(unsubscribed AS INT64)),3) unsubscribe_rate,
       ROUND(AVG(CAST(d1 AS INT64)),3) d1_retention
FROM `persona-496908.<dataset>.v_challenge_metrics_user` -- Δ per-user view
GROUP BY days_completed
ORDER BY days_completed;

-- 3c) IDEAL (if a hold-out or pre-launch cohort exists): compare Challenge-exposed vs not,
-- matched on entry motivation (quiz time_goal / goal). If no hold-out exists, the #1
-- recommendation for the next release is to ship the Challenge as a proper A/B, not 100%.
```

SQL · 10_explore_events

```
-- Stage 1 · probe queries – run these next, paste results back.
-- They tell me the full event list (incl. unsubscribe + any challenge-complete/streak events)
-- and exactly how Challenge participation is logged, so I can build the real tier metrics.

-- Q1 – every event type + volume in app_events
SELECT event_name, COUNT(*) AS n
FROM `persona-496908.sql_assessment.app_events`
GROUP BY event_name
ORDER BY n DESC;

-- Q2 – how is the Challenge logged? (path vs course vs category)
SELECT event_name,
       REGEXP_EXTRACT(path, r'/skills/challenges/(\d+)') AS challenge_id,
       course_id, course_name, category_name, mechanic_name,
       COUNT(*) AS n
FROM `persona-496908.sql_assessment.app_events`
WHERE path LIKE '/skills/challenges/%'
      OR LOWER(event_name) LIKE '%challenge%'
      OR LOWER(IFNULL(category_name, '')) LIKE '%challenge%'
GROUP BY 1,2,3,4,5,6
ORDER BY n DESC
LIMIT 200;

-- Q3 – do challenge lessons carry a distinct course/category? peek at raw challenge rows
SELECT event_name, user_id, timestamp, path, course_id, course_name,
       lesson_id, lesson_name, lesson_order, category_name, mechanic_name, status
FROM `persona-496908.sql_assessment.app_events`
WHERE path LIKE '/skills/challenges/%' OR LOWER(event_name) LIKE '%challenge%'
ORDER BY timestamp
LIMIT 50;

-- Q4 – the unsubscribe event (which event_name carries unsubscribe_reason?)
SELECT event_name, unsubscribe_reason, COUNT(*) AS n
FROM `persona-496908.sql_assessment.app_events`
WHERE unsubscribe_reason IS NOT NULL
GROUP BY 1,2
ORDER BY n DESC
LIMIT 100;
```

SQL · 11_challenge_funnel

```
-- Stage 1 · Part C – first real query. Run this, paste the one-row result back.  
-- A compact engagement funnel using the confirmed event names (Events Convention).  
-- It returns distinct-user counts so we can see the shape before building tier metrics.
```

```
SELECT  
    COUNT(DISTINCT IF(event_name='pr_webapp_launch_first_time',      user_id, NULL)) AS registered,  
    COUNT(DISTINCT IF(event_name='pr_webapp_challenge_view',        user_id, NULL)) AS  
challenge_viewed,  
    COUNT(DISTINCT IF(event_name='pr_webapp_challenge_start',        user_id, NULL)) AS  
challenge_started,  
    COUNT(DISTINCT IF(event_name='pr_webapp_challenge_join',        user_id, NULL)) AS  
challenge_joined,  
    COUNT(DISTINCT IF(event_name='pr_webapp_lesson_completed',      user_id, NULL)) AS  
completed_any_lesson,  
    COUNT(DISTINCT IF(event_name='pr_webapp_challenge_certificate_download',user_id, NULL)) AS  
got_challenge_certificate,  
    COUNT(DISTINCT IF(event_name='pr_webapp_challenge_csat_click',   user_id, NULL)) AS  
gave_challenge_csat,  
    COUNT(DISTINCT IF(event_name='pr_webapp_unsubscribed',          user_id, NULL)) AS unsubscribed,  
    COUNT(DISTINCT user_id) AS all_users,  
    MIN(timestamp) AS first_event, MAX(timestamp) AS last_event  
FROM `persona-496908.sql_assessment.app_events`;
```

```
-- After this: I'll write query 12 = engagement tiers (high/low/didn't-start) and the  
-- D1/D3/D7 + unsubscribe + completion + CSAT comparison, per your definitions.
```


SQL · 12_challenge338_user_table

```
-- 12 · Per-user table for the 7-Day Claude Challenge (id 338).
-- RUN this, then "Save results ► CSV" and drop the file in part-c-release-verdict/data/.
-- One row per user who STARTED challenge 338. I'll do the tier / D1-D3-D7 / segment analysis locally.
-- (If it throws an error, paste me the error text – I can't test against your DB.)

WITH ch338 AS (
    -- every event belonging to challenge 338
    SELECT user_id, event_name, timestamp, DATE(timestamp) AS d, csat_score
    FROM `persona-496908.sql_assessment.app_events`
    WHERE course_id = 338 OR REGEXP_CONTAINS(IFNULL(path,''), r'challenges/338')
),
starter AS (
    -- per-user challenge-338 engagement + outcomes
    SELECT
        user_id,
        MIN(IF(event_name='pr_webapp_challenge_start', d, NULL)) AS start_date,
        COUNT(DISTINCT d) AS active_challenge_days,
        COUNTIF(event_name='pr_webapp_lesson_completed') AS lessons_completed,
        MAX(IF(event_name='pr_webapp_course_completed',1,0)) AS completed_course,
        MAX(IF(event_name IN ('pr_webapp_challenge_certificate_download',
            'pr_webapp_challenge_certificate_share'),1,0)) AS got_certificate,
        COUNTIF(event_name IN ('pr_webapp_lesson_csat_click','pr_webapp_challenge_csat_click')) AS csat_n,
        AVG(IF(event_name IN ('pr_webapp_lesson_csat_click','pr_webapp_challenge_csat_click'),
            csat_score, NULL)) AS avg_csat
    FROM ch338
    GROUP BY user_id
),
act AS ( SELECT user_id, DATE(timestamp) AS d
    FROM `persona-496908.sql_assessment.app_events` GROUP BY 1,2 ),
uns AS ( SELECT user_id, ANY_VALUE(unsubscribe_reason) AS unsub_reason
    FROM `persona-496908.sql_assessment.app_events`
    WHERE event_name='pr_webapp_unsubscribed' GROUP BY 1 ),
seg AS ( SELECT user_id, ANY_VALUE(age) age, ANY_VALUE(gender) gender, ANY_VALUE(goal) goal,
    ANY_VALUE(status) status, ANY_VALUE(subscription) plan
    FROM `persona-496908.sql_assessment.subscribe_events` GROUP BY 1 )
SELECT
    s.user_id, s.start_date, s.active_challenge_days, s.lessons_completed,
    s.completed_course, s.got_certificate, s.csat_n, ROUND(s.avg_csat,2) AS avg_csat,
    MAX(IF(a.d = DATE_ADD(s.start_date, INTERVAL 1 DAY),1,0)) AS ret_d1,
    MAX(IF(a.d = DATE_ADD(s.start_date, INTERVAL 3 DAY),1,0)) AS ret_d3,
    MAX(IF(a.d = DATE_ADD(s.start_date, INTERVAL 7 DAY),1,0)) AS ret_d7,
    IF(u.user_id IS NULL, 0, 1) AS unsubscribed, u.unsub_reason,
    g.age, g.gender, g.goal, g.status, g.plan
FROM starter s
LEFT JOIN act a ON a.user_id = s.user_id
LEFT JOIN uns u ON u.user_id = s.user_id
LEFT JOIN seg g ON g.user_id = s.user_id
WHERE s.start_date IS NOT NULL
-- only users who actually STARTED the challenge
GROUP BY s.user_id, s.start_date, s.active_challenge_days, s.lessons_completed,
    s.completed_course, s.got_certificate, s.csat_n, s.avg_csat,
    u.user_id, u.unsub_reason, g.age, g.gender, g.goal, g.status, g.plan
ORDER BY s.start_date;
```

SQL · 13_comparison_groups

```
-- 13 · THE COMPARISON QUERY – one row per user, takers AND non-takers of Challenge 338.
-- This is the missing third of Part C Task 1: "high / low / didn't take it".
-- Run it once, then "SAVE RESULTS ► CSV (local file)" into part-c-release-verdict/data/.
-- ~10,000 rows (small enough for the local-file download). All tiering + stats done locally in Python.
-- If it errors, paste me the error text verbatim.

WITH
bounds AS (
    -- last day present in the data (for window correction)
    SELECT MAX(DATE(timestamp)) AS last_data_date
    FROM `persona-496908.sql_assessment.app_events`
),
ev AS (
    -- every app event + a flag for "belongs to challenge 338"
    SELECT
        user_id,
        event_name,
        DATE(timestamp) AS d,
        csat_score,
        unsubscribe_reason,
        country_code,
        (course_id = 338 OR REGEXP_CONTAINS(IFNULL(path, ''), r'challenges/338')) AS ch338
    FROM `persona-496908.sql_assessment.app_events`
),
per_user AS (
    -- one row per user: engagement, outcomes, challenge-338
    detail
    SELECT
        user_id,
        MIN(d) AS first_seen,
        COUNT(DISTINCT d) AS active_days_total,
        ANY_VALUE(country_code) AS country_code,

        -- overall product engagement (all courses, not just the challenge)
        COUNTIF(event_name = 'pr_webapp_lesson_completed') AS lessons_completed_all,
        MAX(IF(event_name = 'pr_webapp_onboarding_finished', 1, 0)) AS finished_onboarding,
        MAX(IF(event_name = 'pr_webapp_personal_plan_started', 1, 0)) AS started_personal_plan,

        -- outcome metrics
        MAX(IF(event_name = 'pr_webapp_unsubscribed', 1, 0)) AS unsubscribed,
        ANY_VALUE(IF(event_name = 'pr_webapp_unsubscribed', unsubscribe_reason, NULL)) AS unsub_reason,
        COUNTIF(event_name IN ('pr_webapp_lesson_csat_click',
                                'pr_webapp_module_csat_click',
                                'pr_webapp_challenge_csat_click')) AS csat_n_all,
        AVG(IF(event_name IN ('pr_webapp_lesson_csat_click',
                                'pr_webapp_module_csat_click',
                                'pr_webapp_challenge_csat_click'), csat_score, NULL)) AS avg_csat_all,

        -- exposure to the challenge surface (any challenge – the popup carries no course_id)
        MAX(IF(event_name = 'pr_webapp_challenge_popup_view', 1, 0)) AS saw_challenge_popup,
        MAX(IF(event_name = 'pr_webapp_challenge_popup_click', 1, 0)) AS clicked_challenge_popup,
        MAX(IF(event_name = 'pr_webapp_challenge_view', 1, 0)) AS viewed_any_challenge,
        MAX(IF(event_name = 'pr_webapp_challenge_start', 1, 0)) AS started_any_challenge,

        -- challenge 338 specifically
        MIN(IF(ch338 AND event_name = 'pr_webapp_challenge_start', d, NULL)) AS ch_start_date,
        MAX(IF(ch338 AND event_name = 'pr_webapp_challenge_view', 1, 0)) AS ch_viewed,
        MAX(IF(ch338 AND event_name = 'pr_webapp_challenge_join', 1, 0)) AS ch_joined,
        COUNTIF(ch338 AND event_name = 'pr_webapp_lesson_completed') AS ch_lessons_completed,
        COUNTIF(ch338 AND event_name = 'pr_webapp_lesson_started') AS ch_lessons_started,
        MAX(IF(ch338 AND event_name = 'pr_webapp_course_completed', 1, 0)) AS ch_completed_course,
        MAX(IF(ch338 AND event_name IN ('pr_webapp_challenge_certificate_download',
                                         'pr_webapp_challenge_certificate_share'), 1, 0)) AS ch_certificate,
        COUNTIF(ch338 AND event_name IN ('pr_webapp_lesson_csat_click',
                                         'pr_webapp_challenge_csat_click')) AS ch_csat_n,
```

```

        AVG(IF(ch338 AND event_name IN ('pr_webapp_lesson_csat_click',
                                         'pr_webapp_challenge_csat_click'), csat_score, NULL)) AS ch_avg_csat
FROM ev
GROUP BY user_id
),

anchored AS (
    -- day 0 for retention: challenge start for takers, first app
    day for everyone else
    SELECT
        p.*,
        COALESCE(p.ch_start_date, p.first_seen) AS anchor_date
    FROM per_user p
),

days AS (
    -- distinct active days per user
    SELECT DISTINCT user_id, DATE(timestamp) AS d
    FROM `persona-496908.sql_assessment.app_events`
),

ret AS (
    -- retention off BOTH anchors, so takers and non-takers are
    comparable
    SELECT
        a.user_id,
        MAX(IF(dd.d = DATE_ADD(a.anchor_date, INTERVAL 1 DAY), 1, 0)) AS ret_d1,
        MAX(IF(dd.d = DATE_ADD(a.anchor_date, INTERVAL 3 DAY), 1, 0)) AS ret_d3,
        MAX(IF(dd.d = DATE_ADD(a.anchor_date, INTERVAL 7 DAY), 1, 0)) AS ret_d7,
        MAX(IF(dd.d = DATE_ADD(a.first_seen, INTERVAL 1 DAY), 1, 0)) AS fs_ret_d1,
        MAX(IF(dd.d = DATE_ADD(a.first_seen, INTERVAL 3 DAY), 1, 0)) AS fs_ret_d3,
        MAX(IF(dd.d = DATE_ADD(a.first_seen, INTERVAL 7 DAY), 1, 0)) AS fs_ret_d7
    FROM anchored a
    JOIN days dd USING (user_id)
    GROUP BY a.user_id
),

seg AS (
    -- buyer segmentation (NULL = no purchase row for this user)
    SELECT
        user_id,
        ANY_VALUE(age) AS age,
        ANY_VALUE(gender) AS gender,
        ANY_VALUE(goal) AS goal,
        ANY_VALUE(status) AS status,
        ANY_VALUE(subscription) AS plan,
        MIN(DATE(timestamp)) AS subscribe_date
    FROM `persona-496908.sql_assessment.subscribe_events`
    GROUP BY user_id
)

SELECT
    a.user_id,

    -- the comparison group (I tier "high vs low" locally from ch_lessons_completed)
    CASE
        WHEN a.ch_start_date IS NOT NULL THEN 'took_338'
        WHEN a.ch_joined = 1 THEN 'joined_338_never_started'
        WHEN a.ch_viewed = 1 THEN 'viewed_338_never_joined'
        WHEN a.started_any_challenge = 1 THEN 'took_other_challenge'
        ELSE 'no_challenge'
    END AS group_338,

    a.first_seen,
    a.anchor_date,
    DATE_DIFF(b.last_data_date, a.anchor_date, DAY) AS days_observed, -- <7 ⇒ D7 not observable

    a.ch_start_date, a.ch_lessons_started, a.ch_lessons_completed,
    a.ch_completed_course, a.ch_certificate, a.ch_csat_n, ROUND(a.ch_avg_csat, 2) AS ch_avg_csat,

    a.lessons_completed_all, a.active_days_total,
    a.finished_onboarding, a.started_personal_plan,
    a.saw_challenge_popup, a.clicked_challenge_popup,

```

```
a.viewed_any_challenge, a.started_any_challenge,

r.ret_d1, r.ret_d3, r.ret_d7,
r.fs_ret_d1, r.fs_ret_d3, r.fs_ret_d7,

a.unsubscribed, a.unsub_reason,
a.csat_n_all, ROUND(a.avg_csat_all, 2) AS avg_csat_all,

s.age, s.gender, s.goal, s.status, s.plan, s.subscribe_date,
IF(s.user_id IS NULL, 0, 1) AS has_subscription,
a.country_code
FROM anchored a
CROSS JOIN bounds b
LEFT JOIN ret r ON r.user_id = a.user_id
LEFT JOIN seg s ON s.user_id = a.user_id
ORDER BY a.user_id;
```

Code · Verification — re-derives every published figure

```
"""QA: re-derive every load-bearing number in the write-ups straight from the CSVs and assert it.

If a number in the markdown ever drifts from the data, this fails loudly.
Run: python3 05_qa.py
"""

import csv
import os
from collections import Counter
from lib import load, wilson, two_prop_z, DATA

rows = load()
N = len(rows)
takers = [r for r in rows if r["took"]]
nont = [r for r in rows if not r["took"]]
fails, checks = [], 0

def check(label, actual, expected, tol=0.0005):
    # NB tolerances are deliberately tight: a 0.001 tolerance once hid a 59.557 -> '59.5%' mis-rounding.
    """expected may be an int/float; tol is absolute."""
    global checks
    checks += 1
    ok = abs(actual - expected) <= tol
    if not ok:
        fails.append(f"{label}: computed {actual!r}, document says {expected!r}")
    print(f"  {'ok' if ok else 'FAIL'} {label:<58} {actual:.4f} (doc: {expected})")

def rate(g, k):
    return sum(r[k] for r in g) / len(g)

print("== cohort shape ==")
check("subscribers in cohort", N, 9956, 0)
check("challenge-338 starters", len(takers), 1111, 0)
check("non-takers", len(nont), 8845, 0)
check("all rows have a subscription", sum(r["has_subscription"] for r in rows), 9956, 0)

print("\n== exposure ladder (Part C §1) ==")
check("saw popup", sum(r["saw_challenge_popup"] for r in rows), 3140, 0)
check("saw popup %", sum(r["saw_challenge_popup"] for r in rows) / N, 0.315, 0.0005)
check("clicked popup", sum(r["clicked_challenge_popup"] for r in rows), 2970, 0)
check("viewed a challenge", sum(r["viewed_any_challenge"] for r in rows), 2109, 0)
check("started 338 %", len(takers) / N, 0.112, 0.0005)
check("never reached a surface",
      sum(1 for r in rows if not r["saw_challenge_popup"] and not r["viewed_any_challenge"]), 6716, 0)
check("never reached a surface %",
      sum(1 for r in rows if not r["saw_challenge_popup"] and not r["viewed_any_challenge"]) / N, 0.675,
      0.0005)

print("\n== inside the challenge (Part C §2) ==")
for k, exp in [(1, 612), (2, 308), (3, 158), (8, 47)]:
    check(f"completed >= {k} lessons", sum(1 for r in takers if r["ch_lessons_completed"] >= k), exp, 0)
check("zero lessons %", sum(1 for r in takers if r["ch_lessons_completed"] == 0) / len(takers), 0.449,
      0.0005)
check("lesson1->2 survival", 308 / 612, 0.503, 0.001)
check("completed the course", sum(r["ch_completed_course"] for r in takers), 48, 0)
check("certificate", sum(r["ch_certificate"] for r in takers), 23, 0)

print("\n== tiers + headline table (Part C §4) ==")
high = [r for r in takers if r["ch_lessons_completed"] >= 3]
low = [r for r in takers if r["ch_lessons_completed"] < 3]
```

```

check("HIGH n", len(high), 158, 0)
check("LOW n", len(low), 953, 0)
check("HIGH D1", rate(high, "fs_ret_d1"), 0.696, 0.001)
check("LOW D1", rate(low, "fs_ret_d1"), 0.584, 0.001)
check("DIDNT-TAKE D1", rate(nont, "fs_ret_d1"), 0.217, 0.001)
check("HIGH D3", rate(high, "fs_ret_d3"), 0.570, 0.001)
check("LOW D3", rate(low, "fs_ret_d3"), 0.344, 0.001)
check("DIDNT-TAKE D3", rate(nont, "fs_ret_d3"), 0.100, 0.001)
for label, g, exp, expn in [("HIGH", high, 0.391, 133), ("LOW", low, 0.180, 737), ("DIDNT-TAKE", nont,
0.069, 6345)]:
    obs = [r for r in g if r["fs_days_observed"] >= 7]
    check(f"{label} D7", rate(obs, "fs_ret_d7"), exp, 0.001)
    check(f"{label} D7 denominator", len(obs), expn, 0)
check("HIGH unsub", rate(high, "unsubscribed"), 0.114, 0.001)
check("LOW unsub", rate(low, "unsubscribed"), 0.114, 0.001)
check("DIDNT-TAKE unsub", rate(nont, "unsubscribed"), 0.143, 0.001)

print("\n== the decisive test (Part C §5 Test A) ==")
ctrl = [r for r in rows if r["group_338"] in ("joined_338_never_started", "viewed_338_never_joined")]
check("takers D1", rate(takers, "fs_ret_d1"), 0.600, 0.001)
check("control n", len(ctrl), 994, 0)
check("control D1 (the 59.6%)", rate(ctrl, "fs_ret_d1"), 0.596, 0.0005)
d, z, p = two_prop_z(sum(r["fs_ret_d1"] for r in takers), len(takers),
                      sum(r["fs_ret_d1"] for r in ctrl), len(ctrl))
check("D1 effect (pts)", d, 0.005, 0.001)
check("D1 effect p-value", p, 0.823, 0.005)
d2, _, p2 = two_prop_z(sum(r["unsubscribed"] for r in takers), len(takers),
                      sum(r["unsubscribed"] for r in ctrl), len(ctrl))
check("unsub effect (pts)", d2, -0.023, 0.001)
check("unsub effect p-value", p2, 0.119, 0.005)
check("joined-never-started D1", rate([r for r in rows if r["group_338"] == "joined_338_never_started"],
"fs_ret_d1"), 0.604, 0.001)
check("viewed-never-joined D1", rate([r for r in rows if r["group_338"] == "viewed_338_never_joined"],
"fs_ret_d1"), 0.593, 0.001)

print("\n== immortal time + dose-response (Test B / C) ==")
lag = Counter(r["days_to_challenge"] for r in takers)
check("start on day 0 %", lag[0] / len(takers), 0.239, 0.001)
oneday = [r for r in takers if r["active_challenge_days"] == 1]
hi = [r for r in oneday if r["ch_lessons_completed"] >= 2]
lo = [r for r in oneday if r["ch_lessons_completed"] <= 1]
d3, _, p3 = two_prop_z(sum(r["fs_ret_d1"] for r in hi), len(hi),
                      sum(r["fs_ret_d1"] for r in lo), len(lo))
check("day-0 dose effect (pts, negative)", d3, -0.092, 0.001)
check("day-0 dose p-value", p3, 0.048, 0.002)

print("\n== the mechanic (Test D) ==")
fin = [r for r in takers if r["ch_completed_course"] == 1]
check("finished in 1 day", sum(1 for r in fin if r["active_challenge_days"] == 1), 13, 0)
check("finished in 1 day %", sum(1 for r in fin if r["active_challenge_days"] == 1) / len(fin), 0.271,
0.001)
check("finished in <=2 days %", sum(1 for r in fin if r["active_challenge_days"] <= 2) / len(fin), 0.458,
0.001)
did = [r for r in takers if r["ch_lessons_completed"] >= 1]
check("returned for a 2nd challenge day %",
      sum(1 for r in did if r["active_challenge_days"] >= 2) / len(did), 0.368, 0.001)
check("mean lessons per active challenge day",
      sum(r["ch_lessons_completed"] / max(1, r["active_challenge_days"]) for r in did) / len(did), 1.46,
0.005)

print("\n== the metric that would have fooled the team (§6) ==")
check("takers D1, challenge-anchored", rate(takers, "ret_d1"), 0.398, 0.001)
check("cohort D1, first-app-day", rate([r for r in rows if r["fs_days_observed"] >= 1], "fs_ret_d1"), 0.260,
0.001)

print("\n== CSAT inversion (§8) ==")

def csat(g):

```

```

v = [r["avg_csat_all"] for r in g if r["csat_n_all"] > 0]
return sum(v) / len(v), len(v)

for label, g, exp in [("never took", nont, 3.74), ("LOW", low, 3.53), ("HIGH", high, 3.33), ("finishers",
fin, 3.10)]:
    m, n = csat(g)
    check(f"CSAT {label}", m, exp, 0.005)
chr_ = [r for r in takers if r["ch_csat_n"] > 0]
check("CSAT of challenge-338 content", sum(r["ch_avg_csat"] for r in chr_) / len(chr_), 3.25, 0.005)
check("same users, all content", sum(r["avg_csat_all"] for r in chr_ if r["csat_n_all"] > 0) /
    len([r for r in chr_ if r["csat_n_all"] > 0]), 3.45, 0.005)

print("\n== product-wide baselines (the thesis number) ==")
check("never completed a lesson anywhere %",
    sum(1 for r in rows if r["lessons_completed_all"] == 0) / N, 0.502, 0.001)
check("cohort unsubscribe %", rate(rows, "unsubscribed"), 0.140, 0.001)

print("\n== demographics (Part A + the PM's claim) ==")
check("male %", sum(1 for r in rows if r["gender"] == "Male") / N, 0.597, 0.001)
check("female %", sum(1 for r in rows if r["gender"] == "Female") / N, 0.384, 0.001)
check("aged 45+ (clean buckets) %", sum(1 for r in rows if r["age"] in ("45-54", "55+")) / N, 0.578, 0.001)
check("aged 45+ incl. legacy bucket %",
    sum(1 for r in rows if r["age"] in ("45-54", "55+", "45+")) / N, 0.601, 0.001)
check("men 55+ count", sum(1 for r in rows if r["gender"] == "Male" and r["age"] == "55+"), 2120, 0)
check("men 55+ %", sum(1 for r in rows if r["gender"] == "Male" and r["age"] == "55+") / N, 0.213, 0.001)
check("men 35-54 %", sum(1 for r in rows if r["gender"] == "Male" and r["age"] in ("35-44", "45-54")) / N,
0.249, 0.001)
for a, exp in [("18-24", 0.311), ("25-34", 0.197), ("35-44", 0.145), ("45-54", 0.095), ("55+", 0.097)]:
    check(f"unsub, age {a}", rate([r for r in rows if r["age"] == a], "unsubscribed"), exp, 0.001)
for gl, exp in [("Work faster", 0.109), ("Feel more confident with AI", 0.107), ("Get a quick side hustle",
0.372)]:
    check(f"unsub, goal '{gl}'", rate([r for r in rows if r["goal"] == gl], "unsubscribed"), exp, 0.001)

print("\n== plans (Part D validation) ==")
pc = Counter(r["plan"] for r in rows)
check("1Week share", pc["1Week"] / N, 0.102, 0.001)
check("4Week share incl. special", (pc["4Week"] + pc["4Week_special"]) / N, 0.646, 0.001)
check("12Week share", pc["12Week"] / N, 0.251, 0.001)
check("1Week unsub", rate([r for r in rows if r["plan"] == "1Week"], "unsubscribed"), 0.346, 0.001)
check("4Week unsub", rate([r for r in rows if r["plan"] == "4Week"], "unsubscribed"), 0.122, 0.001)
check("12Week unsub", rate([r for r in rows if r["plan"] == "12Week"], "unsubscribed"), 0.106, 0.001)
blended = (0.102 * 60.42 + 0.646 * 123.70 + 0.251 * 162.15) / 0.999
check("observed-mix blended net LTV", blended, 126.90, 0.02)

print("\n== unsubscribe reasons (§8) ==")
uns = [r for r in rows if r["unsubscribed"]]
check("total unsubscribed", len(uns), 1393, 0)
invol = {"hard_decline", "Mastercard Alert", "dispute", "paypal_refund", "Visa CDRN",
    "fraud", "Merchant visa", "payment_refunded", "fraud_reported"}
ni = sum(1 for r in uns if r["unsub_reason"] in invol)
check("payment/chargeback/dispute", ni, 157, 0)
check("... as % of unsubs", ni / len(uns), 0.113, 0.001)

print("\n== event catalogue (the 'not observable here' claim) ==")
# The verdict rests on primary metrics being uncomputable from this dataset. That is a claim
# about the event catalogue, not about the user table, so it needs its own check or it would
# be the one published number nothing re-derives.
CAT = os.path.join(DATA, "script_job_ee3720964a3e4c3ac51fe64306b65890_0.csv")
cat = {r["event_name"]: int(r["n"]) for r in csv.DictReader(open(CAT))}
check("event catalogue rows (file integrity)", len(cat), 67, 0)
check("subscription_renewed events", cat.get("pr_webapp_subscription_renewed", 0), 17, 0)
check("... per subscriber", cat.get("pr_webapp_subscription_renewed", 0) / N, 0.0017, 0.0001)

print("\n" + "=" * 70)
if fails:
    print(f"{len(fails)} of {checks} CHECKS FAILED:")
    for f in fails:
        print("    -", f)

```

```
    raise SystemExit(1)
print(f"All {checks} checks passed – every documented number matches the data.")
```


Code · Part D — the LTV and A/B model

```
#!/usr/bin/env python3
"""
Part D – Subscription economics model (Jobscape PM take-home).

Task 1: Total Net LTV over a one-year horizon (per plan gross+net, blended).
Task 2: A/B plan-upgrade break-even (4-week -> 12-week upgrade replacing 2nd upsell).

Design decisions / assumptions (documented explicitly, per brief):
- Cohort of 1 subscriber per plan; everyone pays the intro (survival S0 = 1).
- Cmn = P(pay period n | survived to period m). Survival S_k = product of C01..C_{k-1,k}.
- Period 0 = intro (pays Intro Price). Periods 1..12 = recurring (pay Recurring Price),
  weighted by survival. The data supplies exactly 12 renewal transitions (C01..C1112).
- Payment-provider fee = 12% on ALL collected revenue (intro, recurring, upsells). Net = Gross * (1-0.12).
- Upsells are one-time at checkout; expected value = conv_rate * price, applied once to every subscriber.
- HORIZON: two readings are computed because the brief says "one-year horizon" but gives 12
  transitions per plan:
  (A) FULL-CURVE : apply all 12 transitions to every plan (matches the data given; for the
                    4-week plan 13 periods x 4wk = exactly 52 weeks). PRIMARY.
  (B) STRICT-52WK : count only payments landing within 52 calendar weeks
                    (12-week plan -> only 4 recurring periods; others unchanged). SENSITIVITY.
"""

FEE = 0.12

plans = {
    "1-WEEK": dict(mix=0.10, intro=6.93, rec=39.99, rec_wk=4, intro_wk=1,
                  C=[.55,.50,.60,.75,.80,.80,.80,.80,.80,.80,.80,.80],
                  up1=(0.30,1.99), up2=(0.12,0.99)),
    "4-WEEK": dict(mix=0.70, intro=19.99, rec=39.99, rec_wk=4, intro_wk=4,
                  C=[.67,.65,.70,.75,.80,.80,.80,.80,.80,.80,.80,.80],
                  up1=(0.30,69.99), up2=(0.12,29.99)),
    "12-WEEK": dict(mix=0.20, intro=39.99, rec=62.99, rec_wk=12, intro_wk=12,
                  C=[.64,.57,.65,.75,.75,.75,.75,.75,.75,.75,.75,.75],
                  up1=(0.30,69.99), up2=(0.12,29.99)),
}

def survivals(C):
    s, acc = [], 1.0
    for c in C:
        acc *= c
        s.append(acc)      # s[k-1] = survival to recurring period k
    return s

def periods_within_year(intro_wk, rec_wk):
    """Recurring payments CHARGED within the 52-week horizon (strict reading).
    Recurring period k is charged at week intro_wk + (k-1)*rec_wk; count it if that
    charge falls inside the year (LTV = cash collected, so charge date is what counts,
    not whether the service period extends past week 52). Capped at the 12 given transitions.
    -> 12-week plan: charges at wk 12/24/36/48 -> 4 periods; 4-week & 1-week -> 12."""
    n = 0
    for k in range(1, 13):
        if intro_wk + (k - 1) * rec_wk < 52:
            n += 1
        else:
            break
    return n

def recurring_revenue(p, mode):
    s = survivals(p["C"])
    if mode == "full":
        k = 12
    else:
        k = periods_within_year(p["intro_wk"], p["rec_wk"])
    return p["rec"] * sum(s[:k]), k
```

```

def upsell_ev(p):
    return p["up1"][0]*p["up1"][1] + p["up2"][0]*p["up2"][1]

def plan_ltv(p, mode):
    rec_rev, k = recurring_revenue(p, mode)
    sub_gross = p["intro"] + rec_rev
    up = upsell_ev(p)
    gross = sub_gross + up
    return dict(k=k, intro=p["intro"], rec_rev=rec_rev, upsell=up,
               gross=gross, net=gross*(1-FEE))

print("="*72)
print("TASK 1 – LTV per plan (1-year horizon). Fee = 12%.")
for mode, label in [("full", "(A) FULL-CURVE [PRIMARY]"), ("strict", "(B) STRICT-52WK [sensitivity]"]):
    print(f"\n--- Horizon reading {label} ---")
    print(f"{'plan':>8} {'mix':>5} {'recPds':>6} {'intro':>7} {'recRev':>8} {'upsell':>7} {'GROSS':>8} {'NET':>8}")
    blended = 0.0
    for name, p in plans.items():
        r = plan_ltv(p, mode)
        blended += p["mix"]*r["net"]
        print(f"{'name':>8} {p['mix']*100:4.0f}% {r['k']:>6d} {r['intro']:7.2f} {r['rec_rev']:8.2f} {r['upsell']:7.2f} {r['gross']:8.2f} {r['net']:8.2f}")
    print(f"{'BLENDED NET LTV (mix-weighted)':>55}: ${blended:6.2f}")

print("\n"+"="*72)
print("TASK 2 – A/B: 4-week buyer offered Plan Upgrade ($49.99) INSTEAD of 2nd upsell.")
print("Upgrade buyer becomes a 12-week subscriber (12-week recurring economics).")
UP_PRICE = 49.99
for mode, label in [("full", "FULL-CURVE"), ("strict", "STRICT-52WK")]:
    p4, p12 = plans["4-WEEK"], plans["12-WEEK"]
    recRev4, _ = recurring_revenue(p4, mode)
    recRev12, _ = recurring_revenue(p12, mode)
    up2_4 = p4["up2"][0]*p4["up2"][1] # expected 2nd-upsell value being replaced
    # Break-even p*: p*(UP_PRICE + recRev12 - recRev4) = up2_4 (intro & 1st upsell cancel; fee uniform)
    denom = UP_PRICE + recRev12 - recRev4
    p_star = up2_4 / denom
    print(f"\n--- {label} ---")
    print(f" recurring rev 4-week = ${recRev4:6.2f}")
    print(f" recurring rev 12-week = ${recRev12:6.2f} (delta vs 4wk = ${recRev12-recRev4:+.2f})")
    print(f" 2nd-upsell value replaced (0.12*29.99) = ${up2_4:.2f}")
    print(f" each upgrade adds ${denom:.2f} gross; cannibalises ${up2_4:.2f}")
    print(f" >> BREAK-EVEN upgrade conversion p* = {p_star*100:.2f}%")
    # sample scenarios
    ctrl_gross = p4["intro"] + recRev4 + p4["up1"][0]*p4["up1"][1] + up2_4
    print(f" control 4-week gross LTV = ${ctrl_gross:.2f} (net ${ctrl_gross*(1-FEE):.2f})")
    for p in (0.05, 0.10, 0.20, 0.30):
        test_gross = (p4["intro"] + p4["up1"][0]*p4["up1"][1]
                     + (1-p)*recRev4 + p*(UP_PRICE + recRev12))
        lift = (test_gross-ctrl_gross)/ctrl_gross*100
        print(f" p={p*100:4.0f}% test gross LTV ${test_gross:7.2f} net ${test_gross*(1-FEE):7.2f} lift {lift:+5.1f}%")

```

Code · Part A — the segment clustering

```
#!/usr/bin/env python3
"""Part A – derive audience segments from the quiz data instead of asserting them.

WHY THIS EXISTS
    `segment_sizer.py` applied a hand-written if/elif rule chain: the six segments were authored
    from a jobs-to-be-done framework and then counted, with unmatched users falling through to a
    default. That is a taxonomy, not a segmentation – the structure came from the analyst, not the
    data, so it cannot be falsified. This script asks the opposite question: if nobody told the
    data what the segments were, how many groups are actually there, and what are they?

METHOD
    k-modes (Huang, 1998) on the categorical quiz answers.
    · Distance is Hamming – the count of questions two users answered differently.
    · Centroids are per-attribute MODES, not means. k-means is wrong here: a mean over nominal
      levels ("35-44", "Business owner") has no meaning, and one-hot + Euclidean silently
      asserts that every pair of levels is equidistant, which is the assumption under test.
    · Missing answers are kept as their own level. In a 26-step quiz funnel, *not answering*
      is behaviour (where someone dropped), not absent data – discarding it would bias the
      sample toward finishers.
    · k chosen by the elbow in within-cluster cost, cross-checked against interpretability.
    · Multiple restarts per k; the best-cost run wins. Seeded, so it reproduces exactly.

READ THE OUTPUT AS
    cluster profiles are reported by LIFT (share inside the cluster ÷ share in the population),
    not by raw share. The modal answer of a cluster is usually just the modal answer of the whole
    population; what makes a cluster a cluster is what it over-represents.
"""
import collections
import csv
import os
import random
import sys

random.seed(42)
csv.field_size_limit(10 ** 7)

HERE = os.path.dirname(os.path.abspath(__file__))
SRC = os.path.join(HERE, "..", "materials", "quiz-funnel-answers", "Quiz Funnel Answers.csv")

# The questions that carry motive or identity. Chosen for coverage (>25k answers each) and
# because each maps to something a PM could act on. IDs are the `ID` column, not `question_id`
# (which is 0 for every row in this export).
CORE = {
    "391": "age",
    "357": "gender",
    "394": "work_status",
    "331": "learn_claude_for",
    "390": "benefit",
    "332": "ai_experience",
    "336": "career_fear",
    "335": "feels_ready",
    "1241": "used_claude",
    "1426": "field",
}

MIN_ANSWERED = 6          # users answering fewer than this are quiz drop-offs, not segments
NA = "(no answer)"

def load():
    """Pivot the long export (one row per user-question) into one row per user."""
    wide = collections.defaultdict(dict)
    with open(SRC, encoding="utf-8-sig") as f:
        for r in csv.DictReader(f):
            qid = r["ID"]
```

```

        if qid in CORE:
            a = r["question_answer"].strip()
            if a:
                wide[r["epv_user_id"]][CORE[qid]] = a
    cols = list(CORE.values())
    rows = [[u.get(c, NA) for c in cols] for u in wide.values()]
    kept = [r for r in rows if sum(v != NA for v in r) >= MIN_ANSWERED]
    return cols, rows, kept

def cost(rows, modes, assign):
    return sum(sum(a != b for a, b in zip(rows[i], modes[assign[i]])) for i in range(len(rows)))

def kmodes(rows, k, iters=12, restarts=4):
    best = None
    m = len(rows[0])
    for _ in range(restarts):
        modes = [list(r) for r in random.sample(rows, k)]
        assign = [0] * len(rows)
        for _ in range(iters):
            moved = False
            for i, r in enumerate(rows):
                d = [sum(a != b for a, b in zip(r, mo)) for mo in modes]
                new = d.index(min(d))
                if new != assign[i]:
                    assign[i] = new
                    moved = True
            for c in range(k):
                members = [rows[i] for i in range(len(rows)) if assign[i] == c]
                if not members:
                    modes[c] = list(random.choice(rows))
                    continue
                modes[c] = [collections.Counter(mem[j] for mem in members).most_common(1)[0][0]
                           for j in range(m)]
            if not moved:
                break
        c = cost(rows, modes, assign)
        if best is None or c < best[0]:
            best = (c, modes, assign[:])
    return best

def main():
    cols, allrows, rows = load()
    n = len(rows)
    print(f"quiz respondents in export      {len(allrows):,}")
    print(f"answered >= {MIN_ANSWERED} of {len(cols)} core questions  {n:,}  "
          f"({n/len(allrows)*100:.1f}%)  <- the clustering sample\n")

    base = [collections.Counter(r[j] for r in rows) for j in range(len(cols))]

    print("== choosing k: within-cluster Hamming cost ==")
    print("  k      cost    cost/user    improvement")
    prev = None
    results = {}
    for k in range(2, 9):
        c, modes, assign = kmodes(rows, k)
        results[k] = (c, modes, assign)
        imp = f"({prev-c)/prev*100:5.1f}%" if prev else "  -"
        print(f"  {k:>2}  {c:>7,}  {c/n:>8.3f}  {imp}")
        prev = c

    k = int(sys.argv[1]) if len(sys.argv) > 1 else 6
    c, modes, assign = results[k]
    sizes = collections.Counter(assign)
    print(f"\n== k = {k} · cluster profiles (lift = share in cluster / share in population) ==")
    for cl, sz in sizes.most_common():
        members = [rows[i] for i in range(n) if assign[i] == cl]

```

```
print(f"\n — cluster {cl}  n = {sz:}, ({sz/n*100:.1f}%")
marks = []
for j, col in enumerate(cols):
    cnt = collections.Counter(m[j] for m in members)
    for val, v in cnt.most_common(3):
        if val == NA:
            continue
        share = v / sz
        pop = base[j][val] / n
        if pop > 0 and share > .18:
            marks.append((share / pop, col, val, share))
for lift, col, val, share in sorted(marks, reverse=True)[:6]:
    print(f"      {lift:4.2f}x  {share*100:4.1f}%  {col:16} {val[:52]}")

if __name__ == "__main__":
    main()
```